

Executive Summary

Effective exit and risk control rules are as critical as the entry model for AEGIS. Academic and industry research shows that **trading stops** generally improve risk metrics (drawdown, skew) even if they modestly reduce gross returns [11†L358-L365] [39†L297-L305]. Stops mitigate behavioral biases (the disposition effect [39†L346-L350]) and “lock in” losses before they compound. However, their utility depends on market regime: in trending markets they preserve gains, while in mean-reverting or random-walk regimes they can hurt returns [36†L157-L164] [11†L358-L365].

Common stop rules include fixed-percentage stops, volatility/ATR-based stops, trailing stops, time-based exits, gap/overnight stops, and portfolio-level kill-switches. Each has trade-offs: e.g. tight fixed stops cut losses but increase turnover and whipsaws; trailing stops lock in profits but can exit on normal pullbacks [11†L379-L387] [36†L157-L164]. In India’s volatile equity market, stops should adapt to stock volatility: large caps can tolerate tighter stops, while small caps may need wider bands or volatility-adjusted rules. We catalogue these approaches and recommend a framework of candidate policies (P-01...P-08) to simulate.

We propose an **RISK001-A1 Research Plan**: use AEGIS’s historical recommendation database and market data to backtest each policy over the real trade history. Compute metrics (win-rate, avg/median return, MFE/MAE, max drawdown, Sharpe, Ulcer Index, turnover, etc.) [29†L712-L720]. Perform parameter sweeps (e.g. stop levels 5–15%, ATR multipliers 1.5–3×, trailing thresholds, time-horizons) and statistical tests (e.g. compare distributions via t-test/K-S). Decision criteria include a meaningful reduction in tail losses or drawdown with no unacceptable loss of total return. For example, prefer a policy that cuts maximum drawdown in half while sacrificing <5% of CAGR [11†L379-L387].

Monitoring and Governance: All exits will carry machine-readable reason codes (`HARD_STOP`, `TRAILING_STOP`, `GAP_EXIT`, `PORTFOLIO_MAX_DD`, etc.) in the portfolio database. An audit trail logs entry price, stop-level, exit trigger, and exit price. Alerts/alarms should fire if portfolio drawdown exceeds a threshold or if actual performance deviates from backtest beyond a tolerance. We will implement “canary” tests: run the new stops on a shadow portfolio first, compare live vs backtest drift daily, and have rollback rules (e.g. revert to no-stop mode if portfolio underperforms expected drawdowns by >x%).

Next Steps: Immediately begin **RISK001-A1 Exit Analytics** (an analytics notebook), producing `research/policy_comparison.csv` and equity-curve charts for each policy. In parallel, update `docs/RISK001-B_Risk_Controller_Architecture.md` with defined exit conditions and the final risk-controller logic. Short-term (quick-win) actions include introducing a simple fixed stop (e.g. 5–7%) and a maximal drawdown kill-switch into the pipeline. Medium-term, implement best-performing adaptive rule (e.g. ATR or trailing stop) as dictated by RISK001-A1 results. Long-term, explore RL/bandit/meta-learning approaches to fine-tune stops dynamically.

Key Recommendations:

- **Short-term (effort 1–2 days, high impact):** Add a fixed stop (e.g. 7%) and portfolio max-drawdown limit. These immediately cut tail risk.

- **Medium-term (1–2 weeks, moderate impact):** Implement the empirically best static stop (e.g. optimized ATR or trailing rule) from RISK001-A1, plus integrate exit-reason telemetry.
- **Long-term (months, exploratory):** Research ML-driven risk control (reinforcement learning for stop placement, bandit adaptation of stop levels) as part of AEGIS v1.1.

The following sections expand on these points. We provide detailed tables of candidate rules, research methodology, and governance processes. (Appendix lists key papers and blogs.)

1. Literature Survey: Stops & Risk Control in Trading

Stop-Loss Efficacy Depends on Market Dynamics. Early work by Kaminski & Lo (2007) showed that stop losses *only add value in trending environments*. They derive that a stop-loss improves the Sharpe only if the autocorrelation of returns is greater than the strategy's Sharpe [7†L189-L198] [36†L157-L164]. In practice, trending (positive auto-correlation) is needed for a “stopping premium”; in random-walk or mean-reverting cases, stops typically reduce returns [7†L189-L198] [36†L157-L164]. Lo & Remorov (2017) extend this, noting effectiveness also varies by volatility regime.

Empirical Studies: Multiple academic studies in different markets find that judicious stop rules can improve risk-adjusted results. For example, Snorrason & Yusupov (2009) applied fixed and trailing stops on Swedish stocks (1998–2009) and found stop strategies **outperformed buy-and-hold** in both raw and risk-adjusted returns [17†L146-L154]. Similarly, Han (2018) showed that a simple 10% stop on U.S. momentum stocks dramatically cut crashes [16†L175-L184]. Arratia & Dorador (2019) included overnight gaps in their simulations and showed stops helped mitigate gap losses. In contrast, Research Affiliates (2025) tested stop rules on systematic strategies and found no net gain in Sharpe after costs, though **maximum drawdowns were substantially reduced** [11†L358-L365] [11†L379-L387]. Their conclusion: if one cared only about long-run Sharpe, stops are unnecessary, but in practice downside control is often worth a small performance sacrifice [11†L358-L365] [11†L379-L387].

Behavioral Perspective: Beyond the numbers, stop-loss orders have well-documented behavioral benefits. Investors suffer the *disposition effect* (selling winners too early, holding losers too long) [12†L83-L90]. A UK study of retail trades found that using stop-losses “substantially reduced the eagerness to trade gains” and reversed the reluctance to sell losses [12†L83-L90]. Investopedia likewise notes that stops “remove emotion” and act as automatic “safety switches” [39†L297-L305] [39†L346-L350], protecting portfolios especially during volatility.

Industry Practice: Professional CTAs and risk managers widely use stop rules. For instance, CTA consensus guidelines often include multiple layers: single-stock stops (e.g. 4–6% loss), portfolio stops (e.g. 5–10% total drawdown), and use of volatility scaling. CME Group research identifies risk-management “factors” (liquidity, volatility, capacity) and underscores that adjusting positions by these factors can significantly change outcomes [22†L389-L398]. Many systematic funds also incorporate rule-based stop or “escape” conditions to handle black swans. Practitioner blogs and newsletters (e.g. QuantInvesting) echo academic findings that a moderate stop (e.g. 15–20%) often improves results by capping deep losses [15†L25-L34] [39†L297-L305].

Summary: The scholarly consensus is nuanced: **stop-losses can preserve capital** (by reducing max drawdown and tail losses), but at a modest cost to average return. They are most effective in trending markets [36†L157-L164] [11†L379-L387] . For AEGIS (a systematic swing model), preserving capital (Rule #1 in our philosophy) is paramount. Thus, we proceed with a structured evaluation of stop policies, fully aware of the trade-offs from research [11†L358-L365] [39†L346-L350] .

2. Catalog of Stop-Loss Rules

Below we catalogue common stop/exits and their pros/cons, with practical parameter guidelines. Many can be layered together.

- **Fixed % Stop (e.g. P-01: 5% stop):** Exit when position drops by a fixed percentage from entry. *Pros:* Simple and easy to implement. Caps loss at known limit. *Cons:* Too tight stops can trigger on normal noise; too wide may not protect adequately [11†L358-L365] . In U.S. equities, 5–10% is often tested. In volatile Indian mid/small-caps, consider 10–15%. (One could also use separate levels for large vs small caps.)
- **Volatility/ATR Stop (P-02, P-03):** Set stop = entry $\pm k \times \text{ATR}(\text{lookback})$. This adapts to recent volatility. E.g. $\text{ATR}(14) \times 2$ means roughly 2σ move. ATR stops automatically widen in turbulence and tighten in calm. *Pros:* Scales with stock's volatility; avoids uniform stop levels. *Cons:* Requires intraday history or daily ATR. Typical backtest uses 14-day ATR with multipliers 1.5–3× [34†L71-L80] . In trending stocks, the ATR Trailing stop (lock stop to highest price minus $k \times \text{ATR}$) is common [34†L93-L102] .
- **Trailing Stop (P-04):** After entry, move the stop to a fraction (e.g. 5% or $\text{ATR} \times k$) below the highest price reached. *Pros:* Locks in gains (once price moves up, trailing stop rises) and exits on reversals. *Cons:* If market is choppy, might stop out and re-enter many times. Requires holding until price moves up first. For AEGIS's 2M horizon, a trailing 10–20% (or ATR multiple) is plausible. Trailing stops often outperform fixed stops at medium thresholds [16†L169-L174] [36†L160-L164] .
- **Chandelier (High-Water) Stop:** A form of ATR trailing stop using the highest high (or 14-day high) instead of entry. Stop = Highest high – $k \times \text{ATR}$. This keeps the stop further behind in volatile periods. *Pros:* Captures trends more fully; rarely stops out early. *Cons:* More complex; still can be stopped by extreme spikes.
- **Time-Based Exit:** Exit after N days or at end of horizon if neither profit target nor stop hit. *Pros:* Limits holding time on non-performing trades; may capture mean reversion opportunities. *Cons:* Arbitrary, may exit a position that eventually goes big. In AEGIS, one could exit any position after (e.g.) 20 trading days if no target hit. This ensures stale positions don't linger. (No specific citation; common CTA practice.)
- **Break-Even + Trail (P-05):** Once trade is up $x\%$, move stop to entry, then switch to trailing as above. *Pros:* Ensures no loss once a threshold is crossed. *Cons:* Needs careful trigger selection (e.g. +2% or target half). This hybrid is common in discretionary trading.
- **Gap/Emergency Exit (P-06):** If stock gaps down beyond a threshold (e.g. >5–10% drop at open, or hits intraday limit-down), immediately exit at next available price. *Pros:* Protects against news events.

Cons: Hard to test without intraday data; may not be triggered on daily bars. We simulate this by checking if open-of-day is $X\%$ below prior close (or crosses the static stop). *Literature:* Some studies include gap risk in stop evaluations (Arratia & Dorador [36†L165-L168]).

- **Portfolio-Level Stops:**

- **Max Drawdown Kill-Switch (P-07):** If the entire AEGIS portfolio falls by, say, 5% in a day or 10% peak-to-trough, liquidate or suspend trading. *Pros:* Emergency brake on systemic risk. *Cons:* Can trigger during market-wide selloffs (though that's the point). This is akin to market circuit breakers. Rule-of-thumb: set kill-switch at $\sim 1.5\times$ historical max drawdown.
- **Max Position Exposure:** Cap the position size of any single trade (e.g. max 10% of portfolio). Implicit in position sizing, but worth stating.
- **Max Consecutive Losses:** If k losses in a row, tighten stops or take a break. (Discretionary idea; can be implemented as rule.)

Each rule's parameters must be tuned for India's context: Nifty stocks may only move a few % daily, but smallcaps can be erratic. We will test multiple settings via RISK001-A1.

3. Adaptive and ML-Based Approaches

Modern research has explored **adaptive stop rules** beyond fixed formulas:

- **Reinforcement Learning (RL):** RL agents can, in theory, learn optimal exit points by maximizing a risk-adjusted reward. Recent work (Xu et al. 2023; Jiang et al. 2017) applies deep RL to portfolio management, but results are mixed [29†L712-L721] [29†L771-L779] . RL can suffer from excessive complexity: it may learn to avoid positions entirely (zero positions "exploitation"), and typical reward functions omit risk constraints [27†L258-L267] [29†L712-L721] . In practice, RL-based exits would require huge data and are hard to validate. Use case: potentially train an RL policy that chooses a stop distance each day given state (position age, volatility, etc). **Caution:** RL strategies need robust validation and monitoring; they can overfit market history.
- **Multi-Armed Bandits:** We could treat each stop rule (or parameter set) as a "bandit arm" and dynamically allocate more capital to the best performers. For example, run multiple stop policies in parallel (or sub-portfolios) and use bandit algorithms (epsilon-greedy, UCB) to shift allocation to the one yielding highest risk-adjusted returns. This requires enough history and risk budgeting (to avoid blow-up from a "loser" arm). It essentially automates parameter optimization over time.
- **Meta-Learning / Supervised:** One can train a classifier/regressor to predict an optimal stop based on features (volatility, momentum, regime). For instance, a model could learn "if entering when ATR is high and sentiment is bearish, set a tighter stop". However, labeling training examples for "correct stop" is tricky: it's an sequential decision. Some research (Xu et al. 2019) uses tree-based models to predict stop triggers, but these require abundant labeled data and careful feature engineering.
- **Statistical Methods:** An alternative is dynamic rule tweaking: e.g. set stop to maintain constant expected tail loss (CVaR) or use Kelly-like sizing to infer stop levels. For example, one could invert a

parametric distribution of returns to find a stop that limits 99% VaR. This approach is more research than practice in equities.

Risks of ML approaches: They demand much data, are opaque, and may break down in regime shifts. We recommend a cautious approach: first exhaust heuristic rules, then consider simple adaptive schemes. If tried, they must be fully simulated on AEGIS data and monitored via drift detection.

4. Candidate Policies (P-01...P-08)

We propose a menu of stop/exits to test (full definitions below). Each “P-XX” policy is a combination of rules; some have tunable parameters. We will simulate these on AEGIS’s historical trades:

Policy ID	Rules (exit triggers)	Notes/Parameters
P-01 Fixed5	Fixed 5% hard stop from entry; no trailing, no time exit.	Stop=5%; parameter: stop%.
P-02 Fixed7	Fixed 7% hard stop.	Stop=7%.
P-03 ATR(14)*2	Volatility stop: exit when drawdown $\geq 2 \times \text{ATR}(14)$.	ATR period=14d; multiplier=2. Sweep 1.5–3.
P-04 Trailing(10%)	Trailing stop at 10% below peak.	Stops trail top price by 10%. Sweep 5–15%.
P-05 BE+Trail	Exit to breakeven at +2%, then trailing 8% thereafter.	BE trigger=+2%, then trail=8%. Sweep.
P-06 Gap5	Gap/limit-down exit: if open < (prior close – 5%), exit at open.	Gap threshold=5%. (Also apply ATR stop same bar).
P-07 PortDD10	Portfolio drawdown kill-switch: if portfolio down $\geq 10\%$, liquidate all.	(Check daily).
P-08 Hybrid	Combines ATR(1.5 $\times$), trailing 5%, and time-exit 20d.	Mult=1.5 $\times$ ATR; tail=5%; max-hold=20d.

Example simulation rules: For each open position in AEGIS’s history, compute daily holdings. If the stop condition is met on any day (price falls by threshold), mark an exit at that day’s close price (or next open for gap rules). If no stop hits, allow exit at the strategy’s normal exit date or max horizon. For P-07, track portfolio NAV daily; if DD breach, force all open exits at that date.

By sweeping the parameters above (stop% from 3–15%, ATR mult 1–4, trail 3–15%, time 10–60d), we cover most simple variants. In future lab we might add “profit targets” as well, but initial focus is on stops.

5. RISK001-A1 Research Plan (Execution)

Objective: Quantitatively compare the candidate policies above on AEGIS historical data to decide which (if any) improve performance.

Data Inputs:

- `aegis_recommendation_db.csv` (entry date, entry price, exit date, exit price, etc).
- `aegis_registry.csv` (metadata on each recommendation).
- Market data: daily OHLC for all traded symbols (from AEGIS parquet history). Adjust for splits/dividends.
- Scorecard? (for reference, but not needed for stop testing).
- Corporate actions (for adjusted prices).

Computation: For each recommendation and policy:

1. **Reconstruct Position Path:** Starting at entry price, compute the daily equity curve of the position, using daily closes (we assume stop checks at close; for gap rules, compare next open). Track running max price and drawdown.
2. **Identify Exit Point:** Apply rules in chronological order: e.g. for P-01, the exit is the first date D where $(\text{close on D}) \leq (\text{entry}) * (1 - 0.05)$. For trailing, exit when $(\text{close}) \leq (\text{highest close to date}) * (1 - \text{trail}\%)$. For ATR, compute ATR on entry date and each day, exit if drawdown $\geq$ threshold. If no rule triggers, use original exit date or end-of-horizon.
3. **Calculate Metrics (per trade):**

- **Return:** (exit price/entry price - 1).
- **Holding days:** days between entry and exit.
- **MFE (Maximum Favorable Excursion):** max peak above entry before exit.
- **MAE (Max Adverse Excursion):** max drawdown from entry before exit.
- **Time Underwater:** total days spent below entry price.
- **Recovery Time:** for losing trades, days from lowest drawdown back to exit (or entry).

Aggregate for each policy:

- Win Rate (% , profitable trades).
- Average Return, Median Return, Best, Worst.
- Profit Factor ($\text{sum}(\text{winners})/\text{sum}(\text{abs}(\text{losers}))$).
- Sharpe Ratio (mean/std of trade returns).
- Maximum Drawdown of equity curve.
- Ulcer Index (duration/intensity of drawdowns).
- Average Holding Period.
- Turnover (trades per year).

Additionally:

- **Tail Analysis:** Count trades with loss >5%, 10%, 15%, 20%. How many big losses are eliminated by each policy? How many additional losers do they create (e.g. if a winner is stopped out prematurely)?

- **Portfolio Impact:** Recompute portfolio equity curve (assuming trades reinvested or weighted by signal). Compute overall CAGR and drawdown.

Statistical Tests: Compare each policy's return distribution to the baseline (no stops) using paired t-tests or Wilcoxon tests for significance at $p < 0.05$ 【11†L379-L387】. Check if reduction in max drawdown is statistically significant. Use bootstrapping to estimate confidence intervals of metrics.

Decision Criteria: We'll favor any policy that:

- **Significantly lowers maximum drawdown and tail losses**, and
- **Does not significantly harm average/median returns**.

For example: If Policy X reduces max DD by 30% (vs baseline) while cutting CAGR by $< 5\%$, it's likely worth it 【11†L379-L387】. Conversely, a policy that cuts CAGR by 10% to reduce DD by 5% is not justified. The RA analysis suggests we accept a small Sharpe hit for big drawdown gains 【11†L379-L387】. We also review the number of trades and capacity impacts (excessive trading costs may be impractical).

Outputs:

- `research/policy_comparison.csv`: rows for each policy/parameter set, columns for all metrics above.
- `research/equity_curves.csv`: equity curve time series for each policy (for plotting).
- `research/RISK001-A_RESULTS.md`: Narrative report with tables and charts (equity curves, drawdown graphs) summarizing findings, including recommended policy(ies) and parameter values.

Visualization: Plotting equity curves and drawdown charts for baseline vs top policies will clarify trade-offs. (E.g., overlay two equity curves to see tail difference.)

Deliverables: The above files, plus any intermediate data used. Ensure full reproducibility (scripts, seed, parameter records).

6. Monitoring, Telemetry, and Governance

We treat risk-control as a **first-class pipeline component**. Key elements:

- **Machine-Readable Exit Reasons:** For every trade, record a code (e.g. `EXIT_REASON`) such as `HARD_STOP`, `TRAIL_STOP`, `TIME_EXIT`, `GAP_EXIT`, `NORMAL_EXIT`, or `PORTFOLIO_STOP`. Also record parameters at exit time (e.g. stop level hit, days held). This adds columns to the portfolio database and allows post-mortem analysis (e.g. what fraction of exits were by stop, and how did they perform).
- **Audit Trail Fields:** In the daily Telegram report and DB, include: entry date, entry price, exit date, exit price, exit reason, stop price used. The `aegis_today.csv` should include these fields for transparency.

- **Real-Time Alerts:** If any individual stock breaks its stop, an alert could be sent (optional). More importantly, monitor portfolio drawdown in real time: if portfolio drawdown > X% intraday, consider automated action (e.g. pause new entries, tighten all stops). Logging any triggered killswitch events for review is essential.
- **Canary Rollout and Drift Checks:** When deploying a new stop policy, first run it on a “paper” slice or in parallel without affecting real trades. Compare realized vs backtest PnL, drawdowns on a rolling window. If backtest predicted a 3% max drawdown but live hit 10%, that signals model drift. Set thresholds for action (e.g. if live performance deviates >20% from expected, revert to baseline stops).
- **Governance:** Update the strategy versioning to track when stops change (e.g. increment version in Telegram footer). Include governance test coverage: add regression checks ensuring new metrics and reason codes are correctly generated (similar to MON001 fingerprint lock). Any change to exit rules should go through review by risk committee and version-controlled.
- **Metrics Dashboard:** Consider adding dashboard/monitor (OPS002 scope) to track daily: current drawdown, running win/loss stats, average MAE vs stop levels, etc., with alerts for anomalies (e.g. unexpected cluster of stop exits).

By treating the Risk Controller as *binding* (its output overrides the portfolio optimizer), we ensure no stop is ignored. (The optimizer may *recommend* an exit for rotation, but a stop-triggered exit is mandatory.)

7. Implementation Roadmap (Tasks & Prompts)

- **RISK001-A1 Analytics Notebook:** Develop a notebook/script to perform the exit analysis outlined above. *Prompt:* “Implement RISK001-A1 as Python code: read AEGIS DB and OHLC data, apply policies P-01–P-08 over historical trades, compute metrics, output `policy_comparison.csv` and `equity_curves.csv`, and generate `research/RISK001-A_RESULTS.md`.”
- **Policy Sweep Execution:** Run parameter sweeps for each policy (e.g. $\text{stop}\% \in [3,5,7,10]$, $\text{ATR mult} \in [1.5,2,2.5,3]$, $\text{trailing} \in [5,10,15]\%$). Save results to `research/policy_comparison.csv`. Use these to identify “Pareto-optimal” settings (see visual table below).
- **Visualization & Reporting:** Produce tables summarizing top policies (Markdown table or CSV), and charts comparing equity/drawdown curves (to `research/` and embed in markdown). For example:
 - Table: Policy vs Return, Drawdown, etc.
 - Chart: equity curves for Baseline, P-02, P-04 (selected top performers).
 - Chart: distribution of trade returns under each policy.
- **Update Architecture Spec (RISK001-B):** Incorporate any chosen policy into `docs/RISK001-B_Risk_Controller_Architecture.md`. Formalize the exit conditions with pseudo-code or flowcharts. *Prompt:* “Expand RISK001-B doc with finalized risk hierarchy and example state-machine diagram of stop logic.”

- **Governance Tests:** Add automated checks in the regression suite to ensure new fields (e.g. `exit_reason`) appear and obey format. E.g., extend `test_governance.py` to whitelist any new documentation changes, and add `test_ops` to verify the new CSV outputs if needed.
- **Quick Implementation (post-approval):** Once RISK001-A1 identifies the best simple stop, implement it in production code (RISK001-C). Meanwhile, push an interim fix: e.g. add the selected fixed-stop rule (with conservative parameter) in the pipeline to immediately improve tail-risk, subject to rollback as needed.
- **Coordination with OPS002:** As we enhance exits, coordinate with OPS002 tasks (dashboarding, monitoring). For example, add new telemetry fields to MON001 logs.

8. Recommended Visualizations and Tables

To illustrate results and design, we suggest the following visuals:

- **Policy Comparison Table:** A Markdown/CSV table listing each P-01...P-08 (and parameter variations) with key outcomes (Win%, Avg Rtn, MaxDD, Sharpe, etc). This helps compare trade-offs succinctly.
- **Parameter Grid Heatmap:** For each policy, a grid of (e.g.) stop% vs Sharpe or MaxDD, to spot optimal regions.
- **Equity Curve Chart:** Plot the cumulative portfolio value (or cumulative return) over time for Baseline and top candidate policies. E.g. `{ research/equity_curves.csv }` can be used to chart:

```
python -c "import pandas as pd, matplotlib.pyplot as plt;
df=pd.read_csv('research/equity_curves.csv', index_col=0);
df[['Baseline', 'P-02', 'P-04']].cumsum().plot(); plt.savefig('research/
equity_curves.png')"
```

(This will vividly show how drawdowns differ.)

- **Drawdown Chart:** Plot rolling drawdown for baseline vs best stops to highlight reduced peaks.
- **Mermaid Diagrams:**
- **Risk Controller Flowchart:** Shows decision logic, e.g.:

```
graph LR
  A[New Recommendations] --> B{Any Stop Triggered?}
  B -- No --> C[Apply Original Plan]
  B -- Yes --> D[Force Exit at Stop Price]
```

```
C --> E[Execute & Monitor]
D --> E
```

- **State Machine Diagram:** For a single position:

```
stateDiagram-v2
    [*] --> Open
    Open --> Monitoring
    Monitoring --> StopTriggered: price ≤ stop
    Monitoring --> ProfitTaken: profit target
    Monitoring --> TimeExpired: reached max holding
    StopTriggered --> Closed
    ProfitTaken --> Closed
    TimeExpired --> Closed
    Closed --> [*]
```

- **Timeline Diagram:** (optional) showing the daily pipeline or RISK001-A1 workflow steps.

These diagrams (in `docs/RISK001-B_Risk_Controller_Architecture.md`) clarify the design. Equity/drawdown charts (saved as PNG) should be referenced in the final report.

9. Prioritization: Effort vs Impact

Recommendation	Timeframe	Effort	Impact	Rationale
Fixed Stop (5–7%)	Short (1 day)	Low	High	Quick to implement; greatly cuts tail losses 【11†L379-L387】 ; easy to revert if needed.
Max Portfolio DD (10%)	Short (1 day)	Low	High	Forces whole-portfolio exit on crashes; preserves capital in market crashes.
Parameter Sweep (ATR/Trail)	Medium (1 week)	Medium	Medium	Empirical tuning of ATR/Trailing via RISK001-A1. Needs time but yields optimized rule.
Trailing Stop Implementation	Medium (2 weeks)	Medium	Medium	If ATR/trail proves best, code it in. (Also needs monitoring.)
Exit Telemetry & Dashboard	Medium (2 weeks)	Medium	Medium	Adds monitoring for stop events; improves oversight (OPS002 effort overlap).
Adaptive Stops (RL/Bandit)	Long (months)	High	Speculative	Research-phase; potentially high payoff but complex and unproven. Use only after static rules tested.

Recommendation	Timeframe	Effort	Impact	Rationale
Risk Controller Enhancement	Ongoing	High	Long-term	Full productization (RISK001-C) of chosen policies into a robust subsystem.

Short-term quick wins (fixed stops, kill-switch) should be implemented as soon as RISK001-A1 supports them, because they directly address the discovered problem of large losses (current worst loss ~26%). Longer-term efforts include exploring dynamic ML methods and fully integrating risk telemetry (OPS002 and MON002).

Appendix: Key References and Resources

- **Kaminski & Lo (2007)** – “When Do Stop-Loss Rules Stop Losses?” – Seminal paper on stop-loss optimality (trending vs. mean-reversion). Available on SSRN. (Cited in RA [7†L189-L198] .)
- **Lei & Li (2009)** – “The Value of Stop Loss Strategies.” Found trailing stops outperform fixed for stock trading [36†L160-L164] .
- **Lo & Remorov (2017)** – Show stop-loss efficacy depends on serial correlation and regime [36†L160-L164] .
- **Han (2018)** – “Taming Momentum Crashes” – simple stop strategy to cut momentum drawdowns (JFM).
- **Arratia & Dorador (2019)** – Include overnight gap risk in stops.
- **Research Affiliates (2025)** – “Stop the Losses!” – comprehensive whitepaper: no Sharpe gain from stops net-of-cost, but big drawdown reduction [11†L358-L365] [11†L379-L387] .
- **Investopedia – Stop-Loss Orders (2026)** – Good primer: stops as “safety switches” and emotional mitigators [39†L297-L305] [39†L346-L350] .
- **QuantInvesting (2026)** – Blog summarizing empirical stop studies, with findings (10% stop into bonds; trailing stops often best) [16†L67-L74] [16†L169-L174] .
- **LuxAlgo Blog (2025)** – Explanation of ATR-based stops and multipliers [34†L33-L42] [34†L71-L80] . Useful formulas for ATR and trailing stops.
- **CME Group (2017)** – “Quantifying CTA Risk Management” – CTA perspective on risk factors (vol, liquidity, etc) [22†L389-L398] .
- **City, University of London (2012)** – “Stoplosses and the Disposition Effect” (UK dataset): shows stop-losses reduce disposition bias [12†L83-L90] .
- **Investopedia – Circuit Breakers (NSE)** – Explains India’s 10/15/20% market-wide halts (for background on extreme volatility handling).

These (and others cited above) provide both theoretical insight and practical benchmarks for selecting and tuning stop rules. They form the foundation for our RISK001-A1 analysis and architecture design.